

# HealthConnect Appointment Analytics

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## Final Analytics, Dashboard & Business Insights

Data Analytics Track • AnalystLab Africa Experience Lab

- 5,000 appointment records analysed
- Power BI dashboard refined and tested
- Final KPIs, recommendations and showcase materials prepared

FINAL OBJECTIVE

**Show how validated analytics supports HealthConnect decisions to reduce missed appointments and improve appointment attendance.**

Final output

Power BI dashboard, final analytics report, executive summary and presentation materials.

Business value

Clear KPIs, attendance patterns and actionable recommendations for operational decision support.

# 1. Business Problem & Analytics Role

From appointment data to practical decision support

PROBLEM

**HealthConnect needs to reduce missed appointments, improve appointment attendance, make better use of appointment slots, and strengthen administrative support.**

- Missed appointments affect clinic efficiency and patient support.
- Decision-makers need reliable KPIs and attendance-driver patterns.
- The dashboard turns appointment data into a reusable monitoring tool.

MY ANALYTICS COMPONENT

| Data                      | Tools                          | Output                                    | Use                                    |
|---------------------------|--------------------------------|-------------------------------------------|----------------------------------------|
| 5,000 appointment records | Power BI<br>Power Query<br>DAX | Overview and Attendance Drivers dashboard | Decision support for no-show reduction |

**Final Week 8 focus: present a polished, tested, business-focused decision-support package.**

## 2. Final KPI Set

Confirmed metrics from the tested Power BI dashboard

51.2%

No-show rate

Main operational concern

5.3%

Cancellation rate

Cancelled appointments

72.7%

Reminder coverage

Reminder process reach

57.8%

Repeat no-show rate

Previous attendance risk

### VALIDATION EVIDENCE

Week 7 testing confirmed that the KPI values remained consistent. The Year-Month and Appointment Type slicers updated the dashboard correctly and returned to the baseline values when cleared.

Evidence files: KPI checks, slicer screenshots, refinement screenshot, retest screenshots and testing record.

# 3. Validated Findings & Attendance Patterns

What the data revealed about appointment behaviour

## KEY FINDINGS

- No-show behaviour is a major operational issue.
- Previous no-show history remains an important risk dimension.
- Reminder coverage is useful as a process improvement metric.
- Lead time, appointment type and appointment time help segment attendance behaviour.

## DECISION-SUPPORT LOGIC

### 1. Measure

Use final KPIs to confirm the scale of missed appointments.

### 2. Segment

Use attendance drivers to locate where risk patterns may be concentrated.

### 3. Act

Use recommendations to guide reminders, follow-up and monitoring.

**Interpretation: the dashboard helps HealthConnect move from a single no-show rate to targeted operational decisions.**

# 4. Testing, Refinement & Final Validation

How the dashboard was tested and improved before the final showcase

TESTS COMPLETED

KPI consistency

Baseline KPIs displayed correctly: 51.2%, 5.3%, 72.7%, 57.8%.

Year-Month slicer

April filter updated visuals and returned to baseline after clearing.

Appointment Type slicer

General Consultation filter updated visuals and returned to baseline after clearing.

Refinement retest

Validated Key Findings section added; April and General Consultation retests passed.

REFINEMENT MADE

Added a “Validated Key Findings” section to improve dashboard interpretation and make confirmed results easier to communicate.

FINAL VALIDATION OUTCOME

Retesting confirmed that the dashboard remained functional after the refinement.

# 5. Recommendations & Business Value

Translating the final analytics into HealthConnect actions

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## ACTIONABLE RECOMMENDATIONS

- Prioritise patients with previous no-show history for extra follow-up.
- Improve reminder coverage and monitor reminder gaps.
- Use additional reminder touchpoints for longer lead-time bookings.
- Review no-show patterns by appointment type and month.
- Use the dashboard regularly for operational monitoring.

## BUSINESS VALUE

- Focuses attention on higher-risk appointment groups.
- Supports better use of appointment slots.
- Creates a tested dashboard for ongoing monitoring.
- Provides validated findings for modelling and project decisions.

**Final value statement: the analytics component helps HealthConnect understand where appointment attendance interventions should be targeted.**

# 6. Final Integration, Limitations & Closing

How the analytics contribution fits into the wider HealthConnect solution

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## FINAL INTEGRATION POINT

### **Data Analytics → Data Science / Project Management**

Validated no-show patterns and KPIs can support model interpretation, project recommendations and final solution decisions.

*Note: insert actual cross-track evidence after the exchange is completed.*

## LIMITATIONS

- Analysis is descriptive, not causal.
- Cross-track evidence must be added once completed.
- Further modelling and testing can strengthen future versions.

## CLOSING TAKEAWAY

**The final analytics component gives HealthConnect a tested, presentation-ready decision-support dashboard for monitoring missed appointments and guiding targeted improvements.**